

M1104

Partial



20
x/6

1
x 5
2/3

17



Exercise I.(course)

[6 marks]

Let f and g be two bounded and integrable functions on an interval $[a, b]$. Show that if g is of constant sign on $[a, b]$, then there exists a real k such that :

$$k \in \left[\inf_{x \in [a, b]} f(x), \sup_{x \in [a, b]} f(x) \right] \text{ and } \int_a^b f(x)g(x) dx = k \int_a^b g(x) dx.$$

Exercise II.

[6 marks]

Let $f : [a, b] \mapsto \mathbb{R}$ be a continuous function such that $f(a + b - x) = f(x)$ for all $x \in [a, b]$.

1. Show that $\int_a^b x f(x) dx = \frac{a+b}{2} \int_a^b f(x) dx$.
2. Deduce the value of $M = \int_0^{\frac{\pi}{2}} x \frac{\sin(2x)}{1 + \cos^2(2x)} dx$.

Exercise III.

[8 marks]

1. Evaluate the following integrals :

a. $I(x) = \int (1 + \tan x)^3 \ln(1 + \tan x)(1 + \tan^2 x) dx$.

b. $J(x) = \int \frac{-2x + 5}{\sqrt{-x^2 + 4x + 5}} dx$.

2. Evaluate $K(t) = \int \frac{dt}{(t - (1 + \sqrt{2}))(t - (1 - \sqrt{2}))}$.

Deduce the value of $L = \int_0^{\frac{\pi}{2}} \frac{1}{\cos x + \sin x} dx$.

Exercise IV.

[5 marks]

Let $n \in \mathbb{N}$ and $I_n(x) = \int x^n \sqrt{1-x} dx$.

1. Calculate $I_0(x)$.
2. Show that $\int x^{n-1}(1-x)^{\frac{3}{2}} dx = I_{n-1}(x) - I_n(x)$.
3. Using integration by parts, find a relation between $I_n(x)$ and $I_{n-1}(x)$ for $n \geq 1$.

Exercise V.

[5 marks]

Let $u_n = \frac{1}{n\sqrt{n}} \sum_{k=1}^n E(\sqrt{k})$, where $E(x)$ denotes the integer part of the real x .

1. Calculate $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \sqrt{\frac{k}{n}}$. Deduce $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \left(\sqrt{\frac{k}{n}} - \frac{1}{\sqrt{n}} \right)$.

2. Show that $\frac{1}{n} \sum_{k=1}^n \left(\sqrt{\frac{k}{n}} - \frac{1}{\sqrt{n}} \right) < u_n \leq \frac{1}{n} \sum_{k=1}^n \sqrt{\frac{k}{n}}$.

Hint : You may use $x - 1 < E(x) \leq x$ for any real number x .

3. Deduce the value of $\lim_{n \rightarrow \infty} u_n$.



Exercise I. (course)

[6 marks]

Let f and g be two bounded and integrable functions on an interval $[a, b]$. Show that if g is of constant sign on $[a, b]$, then there exists a real k such that :

$$k \in \left[\inf_{x \in [a, b]} f(x), \sup_{x \in [a, b]} f(x) \right] \text{ and } \int_a^b f(x)g(x) dx = k \int_a^b g(x) dx.$$

Solution.

We have f is a bounded function on $[a, b]$, then $m := \inf_{x \in [a, b]} f(x)$ and $M := \sup_{x \in [a, b]} f(x)$ exist and $m \leq f(x) \leq M$ for all $x \in [a, b]$. Assume $g(x) \geq 0$ for all $x \in [a, b]$.

$$m \leq f(x) \leq M \Rightarrow mg(x) \leq f(x)g(x) \leq Mg(x), (\because g(x) \geq 0)$$

$$\Rightarrow m \int_a^b g(x) dx \leq \int_a^b f(x)g(x) dx \leq M \int_a^b g(x) dx. (\because a < b, g \text{ and } f \cdot g \text{ are integrable on } [a, b].)$$

We discuss two cases :

- If $\int_a^b g(x) dx = 0$, then $\int_a^b f(x)g(x) dx = 0$. In this case k can be chosen arbitrary in $[m, M]$ so that

the equality $\int_a^b f(x)g(x) dx = k \int_a^b g(x) dx$ holds true.

- If $\int_a^b g(x) dx > 0$, then $m \leq \frac{\int_a^b f(x)g(x) dx}{\int_a^b g(x) dx} \leq M$. Let $k = \frac{\int_a^b f(x)g(x) dx}{\int_a^b g(x) dx}$, then $k \in [m, M]$

and the equality $\int_a^b f(x)g(x) dx = k \int_a^b g(x) dx$ holds true.

Exercise II.

[6 marks]

Let $f : [a, b] \mapsto \mathbb{R}$ be a continuous function such that $f(a + b - x) = f(x)$ for all $x \in [a, b]$.

$$1. \text{ Show that } \int_a^b xf(x) dx = \frac{a+b}{2} \int_a^b f(x) dx.$$

$$2. \text{ Deduce the value of } M = \int_0^{\frac{\pi}{2}} x \frac{\sin(2x)}{1 + \cos^2(2x)} dx.$$

Solution.

1. Let $u = a + b - x$. We have u is of class C^1 on $[a, b]$, ($\because$ it is polynomial), $f(u)$ is continuous on $[a, b]$ and $dx = -du$. Hence,

$$\begin{aligned} \int_a^b xf(x) dx &= \int_b^a (a+b-u)f(a+b-u)(-du) = \int_a^b (a+b-u)f(u) du \\ &= \int_a^b (a+b)f(u) du - \int_a^b uf(u) du = \int_a^b (a+b)f(x) dx - \int_a^b xf(x) dx. \text{ so,} \end{aligned}$$

$$2 \int_a^b xf(x) dx = (a+b) \int_a^b f(x) dx. \text{ Therefore } \int_a^b xf(x) dx = \frac{a+b}{2} \int_a^b f(x) dx.$$

2. Let $f(x) = \frac{\sin(2x)}{1 + \cos^2(2x)}$. We have :

- f is continuous on $[0, \frac{\pi}{2}]$, because it is the quotient of continuous functions with denominator never vanishes on $[0, \frac{\pi}{2}]$.

- $f(0 + \frac{\pi}{2} - x) = f(x)$ for all $x \in [0, \frac{\pi}{2}]$. Using part 1., we obtain

$$\begin{aligned} M &= \int_0^{\frac{\pi}{2}} x \frac{\sin(2x)}{1 + \cos^2(2x)} dx = \frac{0 + \frac{\pi}{2}}{2} \int_0^{\frac{\pi}{2}} \frac{\sin(2x)}{1 + \cos^2(2x)} dx \\ &= \frac{\pi}{4} \int_1^{-1} \frac{-\frac{1}{2} du}{1 + u^2}; (\text{ We take } \cos 2x = u \in C^1([0, \frac{\pi}{2}]) \text{ and } g(u) = \frac{-\frac{1}{2}}{1 + u^2} \text{ continuous on } [-1, 1].) \\ &= \frac{\pi}{8} \int_{-1}^1 \frac{du}{1 + u^2} = \frac{\pi}{8} (\arctan 1 - \arctan -1) = \frac{\pi^2}{16}. \end{aligned}$$



Exercise III.

[8 marks]

1. Evaluate the following integrals :

a. $I(x) = \int (1 + \tan x)^3 \ln(1 + \tan x)(1 + \tan^2 x) dx.$

b. $J(x) = \int \frac{-2x + 5}{\sqrt{-x^2 + 4x + 5}} dx.$

2. Evaluate $K(t) = \int \frac{dt}{(t - (1 + \sqrt{2}))(t - (1 - \sqrt{2}))}.$

Deduce the value of $L = \int_0^{\frac{\pi}{2}} \frac{1}{\cos x + \sin x} dx.$

Solution. 1.

$$I(x) = \int t^3 \ln t dt; \text{ (We take } 1 + \tan x = t, \text{ hence } (1 + \tan^2 x)dx = dt.)$$

$$= \frac{t^4}{4} \ln t - \int \frac{t^4}{4} \cdot \frac{1}{t} dt; \text{ (By parts, we take } U' = t^3 \text{ and } V = \ln t.)$$

$$= \frac{t^4}{4} \ln t - \int \frac{t^3}{4} dt = \frac{t^4}{4} \ln t - \frac{t^4}{16} + c = \frac{(1 + \tan x)^4}{4} \ln(1 + \tan x) - \frac{(1 + \tan x)^4}{16} + c.$$

$$J(x) = \int \frac{-2x + 4 + 1}{\sqrt{-x^2 + 4x + 5}} dx = \int \frac{-2x + 4}{\sqrt{-x^2 + 4x + 5}} dx + \int \frac{1}{\sqrt{-x^2 + 4x + 5}} dx$$

$$= \int \frac{d(-x^2 + 4x + 5)}{\sqrt{-x^2 + 4x + 5}} + \int \frac{d(x - 2)}{\sqrt{9 - (x - 2)^2}}$$

$$= 2\sqrt{-x^2 + 4x + 5} + \arcsin\left(\frac{x - 2}{3}\right) + c.$$

2. We decompose the integrand into simple elements.

$$\frac{1}{(t - (1 + \sqrt{2}))(t - (1 - \sqrt{2}))} = \frac{a}{(t - (1 + \sqrt{2}))} + \frac{b}{(t - (1 - \sqrt{2}))}$$

$$\Rightarrow 1 = a(t - (1 - \sqrt{2})) + b(t - (1 + \sqrt{2})); \text{ (Take } t = (1 + \sqrt{2}) \text{ and } t = (1 - \sqrt{2}).)$$

$$\Rightarrow a = \frac{1}{2\sqrt{2}} \text{ and } b = \frac{-1}{2\sqrt{2}}.$$

$$\text{Hence, } K(t) = \frac{1}{2\sqrt{2}} \ln |t - (1 + \sqrt{2})| - \frac{1}{2\sqrt{2}} \ln |t - (1 - \sqrt{2})| + c = \frac{1}{2\sqrt{2}} \ln \left| \frac{t - (1 + \sqrt{2})}{t - (1 - \sqrt{2})} \right| + c.$$

$$L = \int_0^{\frac{\pi}{2}} \frac{1}{\cos x + \sin x} dx; \quad (t = \tan \frac{x}{2} \in C^1([0, \frac{\pi}{2}])) \text{ because } \frac{x}{2} \in [0, \frac{\pi}{4}] \text{ for } x \in [0, \frac{\pi}{2}].)$$

$$= \int_0^1 \frac{\frac{2dt}{1+t^2}}{\frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2}} = -2 \int_0^1 \frac{dt}{t^2 - 2t - 1} = -2 \int_0^1 \frac{dt}{(t - (1 + \sqrt{2}))(t - (1 - \sqrt{2}))}$$

$$= -2(k(1) - k(0)) = 2k(0) = \sqrt{2} \ln(\sqrt{2} + 1).$$

Exercise IV.

[5 marks]

Let $n \in \mathbb{N}$ and $I_n(x) = \int x^n \sqrt{1-x} dx.$ 1. Calculate $I_0(x).$ 2. Show that $\int x^{n-1}(1-x)^{\frac{3}{2}} dx = I_{n-1}(x) - I_n(x).$ 3. Using integration by parts, find a relation between $I_n(x)$ and $I_{n-1}(x)$ for $n \geq 1.$

Solution.

$$1. I_0(x) = \int \sqrt{1-x} dx = - \int \sqrt{1-x} d(1-x) = -\frac{2}{3}(1-x)^{\frac{3}{2}} + c.$$

$$2. \int x^{n-1}(1-x)^{\frac{3}{2}} dx = \int x^{n-1}(1-x)(1-x)^{\frac{1}{2}} dx = \int (x^{n-1} - x^n)(1-x)^{\frac{1}{2}} dx = I_{n-1}(x) - I_n(x).$$

$$3. I_n(x) = \int x^n \sqrt{1-x} dx; \text{ (By parts, we take } U' = \sqrt{1-x} \text{ and } V = x^n. \text{ So } U = -\frac{2}{3}(1-x)^{\frac{3}{2}} \text{ and } V' = nx^{n-1}).$$

$$= -\frac{2}{3}(1-x)^{\frac{3}{2}}x^n + n \frac{2}{3} \int (1-x)^{\frac{3}{2}} x^{n-1} dx = -\frac{2}{3}(1-x)^{\frac{3}{2}}x^n + \frac{2}{3}n(I_{n-1}(x) - I_n(x)). \text{ so,}$$

$$\left(\frac{2}{3}n+1\right)I_n(x) = -\frac{2}{3}(1-x)^{\frac{3}{2}}x^n + \frac{2}{3}nI_{n-1}(x).$$

Exercise V.

[5 marks]

Let $u_n = \frac{1}{n\sqrt{n}} \sum_{k=1}^n E(\sqrt{k})$, where $E(x)$ denotes the integer part of the real x .

1. Calculate $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \sqrt{\frac{k}{n}}$. Deduce $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \left(\sqrt{\frac{k}{n}} - \frac{1}{\sqrt{n}} \right)$.
2. Show that $\frac{1}{n} \sum_{k=1}^n \left(\sqrt{\frac{k}{n}} - \frac{1}{\sqrt{n}} \right) < u_n \leq \frac{1}{n} \sum_{k=1}^n \sqrt{\frac{k}{n}}$.
Hint : You may use $x - 1 < E(x) \leq x$ for any real number x .
3. Deduce the value of $\lim_{n \rightarrow \infty} u_n$.

Solution.

$$1. \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \sqrt{\frac{k}{n}} = \int_0^1 \sqrt{x} dx = \frac{2}{3}; \text{ (Since } f(x) = \sqrt{x} \text{ is continuous, hence integrable, on } [0, 1]. \text{)}$$

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \left(\sqrt{\frac{k}{n}} - \frac{1}{\sqrt{n}} \right) &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \sqrt{\frac{k}{n}} - \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \frac{1}{\sqrt{n}} = \frac{2}{3} - \lim_{n \rightarrow \infty} \frac{1}{n\sqrt{n}} \sum_{k=1}^n 1 \\ &= \frac{2}{3} - \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} = \frac{2}{3}. \end{aligned}$$

$$\begin{aligned} 2. \sqrt{k} - 1 < E(\sqrt{k}) \leq \sqrt{k} &\Rightarrow \sum_{k=1}^n (\sqrt{k} - 1) < \sum_{k=1}^n E(\sqrt{k}) \leq \sum_{k=1}^n \sqrt{k} \\ &\Rightarrow \frac{1}{n\sqrt{n}} \sum_{k=1}^n (\sqrt{k} - 1) < \frac{1}{n\sqrt{n}} \sum_{k=1}^n E(\sqrt{k}) \leq \frac{1}{n\sqrt{n}} \sum_{k=1}^n \sqrt{k}; \text{ (Since } \frac{1}{n\sqrt{n}} > 0. \text{)} \\ &\Rightarrow \frac{1}{n} \sum_{k=1}^n \left(\sqrt{\frac{k}{n}} - \frac{1}{\sqrt{n}} \right) < u_n \leq \frac{1}{n} \sum_{k=1}^n \sqrt{\frac{k}{n}}. \end{aligned}$$

$$3. \lim_{n \rightarrow \infty} u_n = \frac{2}{3}.$$



Exercise 1 (07 points)

1. (Course) Let I be an interval of $\mathbb{R}$ such that $\forall x \in I, -x \in I$, and f a continuous function on I .

(a) Prove that if f is even, then, $\forall x \in I, \int_{-x}^x f(t)dt = 2 \int_0^x f(t)dt$.

(b) Prove that if f is odd, then, $\forall x \in I, \int_{-x}^x f(t)dt = 0$.

2. Calculate $\int_0^{2\pi} \frac{(\sin x)^2 \sin(2x)}{1 + (\sin x)^8} dx$.

Exercise 2 (16 points)

1. Evaluate the following integrals:

$$I(x) = \int \frac{dx}{x(\ln x)^2}, \quad J(x) = \int \frac{x}{\sqrt{x+2}} dx, \quad K(x) = \int \frac{\cos x}{\cos x + \sin x} dx.$$

2. Evaluate $L(x) = \int \frac{2x}{x^2 + x + 4} dx$. Deduce $M(x) = \int \frac{\cos x \sin x}{(\cos x)^2 + \cos x + 4} dx$.

3. Give the general form of the partial fraction decomposition of the function

$$f(x) = \frac{x^2 - 1}{(x^2 - 4)^2(x^2 + 4)^2}$$

without calculating the constants.

Exercise 3 (07 points) Let $n \in \mathbb{N}$ and $I_n = \int_0^\pi (\sin x)^{2n} dx$.

a) Calculate I_0 and I_1 .

b) Using integration by parts, find a relation between I_n and I_{n-1} for $n \geq 2$.



Exercise I. [7 marks]

1. Given I interval of $\mathbb{R}$, centered at zero & f cont on I .a) f even $\Rightarrow \forall x \in I, \int_{-x}^x f(t) dt = 2 \int_0^x f(t) dt$?

$$\int_{-x}^x f(t) dt \stackrel{(1)}{=} \int_{-x}^0 f(t) dt + \int_0^x f(t) dt \quad \left| \begin{array}{l} \text{change relation:} \\ \text{since } f \text{ integrable (}\neq \text{cont) on } [-x, x] \end{array} \right.$$

$$= I_1 + I_2$$

$$I_1 = \int_{-x}^0 f(t) dt \quad \text{take } t = -u, \quad dt = -du$$

$$\stackrel{(1/2)}{=} \int_x^0 f(-u) (-du) \stackrel{(1/2)}{=} - \int_x^0 f(u) du \quad \text{since } f \text{ even}$$

$$= \int_0^x f(u) du = I_2$$

$$\text{Hence } \int_{-x}^x f(t) dt = I_1 + I_2 = 2 I_2 \stackrel{(1/2)}{=} 2 \int_0^x f(t) dt.$$

b) f odd $\Rightarrow \forall x \in I, \int_{-x}^x f(t) dt = 0$?

$$\int_{-x}^x f(t) dt = \int_{-x}^0 f(t) dt + \int_0^x f(t) dt = I_1 + I_2$$

$$I_1 = \int_{-x}^0 f(t) dt \quad \text{take } t = -u, \quad dt = -du$$

$$= - \int_x^0 f(-u) du \stackrel{(1/2)}{=} - \int_x^0 f(u) du \quad \text{since } f \text{ odd}$$

$$\stackrel{(1/2)}{=} - \int_0^x f(u) du = -I_2$$

$$\text{Hence } \int_{-x}^x f(t) dt = -I_2 + I_2 \stackrel{(1/2)}{=} 0$$

$$2. \text{ Calculate } I = \int_0^{2\pi} \frac{(\sin x)^2}{1 + (\sin x)^8} dx = \int f(x) dx.$$

$$f \stackrel{(1/2)}{\text{is odd}}, \pi\text{-periodic \& cont}, I = \int_0^{2\pi} f(x) dx \stackrel{(1/2)}{=} \int_{-\pi}^{\pi} f(x) dx \stackrel{(1/2)}{=} 0$$

2nd way:

B. Rule: $u(x) = \sin x$ of class C^1 on $[0, 2\pi]$.

$u([0, 2\pi]) = [-1, 1]$

$f(u) = \frac{2u^3}{1+u^8}$ is cont on $[-1, 1] = u([0, 2\pi])$,

$I = \int_0^{2\pi} \frac{2u^2 u}{1+u^8} du = \int_{-1}^1 \frac{2u^2}{1+u^8} du$

or you can take $u(x) = \cos x$

or $u(x) = \sin^2 x$

or $u(x) = \cos^2 x$

or $u(x) = \sin^4 x$

Exercise 3. [7 marks]

$I_n = \int_0^\pi (\sin x)^{2n} dx \quad n \in \mathbb{N}$

a) $I_0 = \int_0^\pi dx = \pi$ (1)

$I_1 = \int_0^\pi \sin^2 x dx = \frac{1}{2} \int_0^\pi (1 - \cos 2x) dx$
 $= \frac{1}{2} [x - \frac{1}{2} \sin 2x]_0^\pi = \frac{\pi}{2}$

b) $I_n = \int_0^\pi \underbrace{\sin x}_{u'} \underbrace{(\sin x)^{2n-1}}_v dx$
 $\begin{cases} u' = \sin x \Rightarrow u = -\cos x \\ v = (\sin x)^{2n-1} \Rightarrow v' = (2n-1)(\sin x)^{2n-2} \cos x \end{cases}$
 $= [-\cos x (\sin x)^{2n-1}]_0^\pi - \int_0^\pi (-\cos x) (2n-1) (\sin x)^{2n-2} \cos x dx$
 $= (2n-1) \int_0^\pi \cos^2 x (\sin x)^{2n-2} dx$
 $= (2n-1) \int_0^\pi (1 - \sin^2 x) (\sin x)^{2n-2} dx$
 $= (2n-1) [I_{n-1} - I_n]$
 $\Rightarrow I_n = \frac{2n-1}{2n} I_{n-1}$

$$J(x) = \int \frac{x}{\sqrt{x+2}} dx = 2x \sqrt{x+2} - 2 \int \sqrt{x+2} dx$$

$$= 2x \sqrt{x+2} - \frac{4}{3} (x+2)^{3/2}$$

practical method

Exercise II. [16 marks]

$$1. \rightarrow I(x) = \int \frac{dx}{x(\ln x)^2} = \int \frac{d(\ln x)}{(\ln x)^2} = -\frac{1}{\ln x} + C$$

$$\rightarrow J(x) = \int \frac{x+2}{\sqrt{x+2}} dx = \int \sqrt{x+2} dx \Rightarrow x+2 = t^2 \Rightarrow dx = 2t dt$$

$$= \int \frac{(t^2-2)}{t} 2t dt = 2 \int (t^2-2) dt = \frac{2}{3} t^3 - 4t + C = \frac{2}{3} \sqrt{x+2}^3 - 4\sqrt{x+2} + C$$

$$\rightarrow K(x) = \int \frac{\cos x}{\cos x + \sin x} dx$$

$$= \int \frac{\cos^2 x}{\cos^2 x + \sin^2 x} dx = \int \frac{\cos^2 x}{1} dx$$

$$= \int \frac{1}{1+t^2} dt = \int \frac{dt}{(1+t)(1+it)}$$

B.R. Rule: $\frac{1}{2}$

$t = \tan x \Rightarrow dt = (1+\tan^2 x) dx = \frac{1}{\cos^2 x} dx$

$x = \arctan t$

Decomposition

$$\frac{1}{(1+t)(1+t^2)} = \frac{A}{1+t} + \frac{Bt+C}{1+t^2}$$

$$\rightarrow 1 = A(1+t^2) + (Bt+C)(1+t)$$

$$t = -1 \Rightarrow 1 = 2A \Rightarrow A = \frac{1}{2}$$

$$t = 0 \Rightarrow 1 = \frac{1}{2}(1) + C \Rightarrow C = \frac{1}{2}$$

$$t = 1 \Rightarrow 1 = \frac{1}{2}(2) + (B+\frac{1}{2})2 \Rightarrow B = -\frac{1}{2}$$

Hence,

$$K(x) = \int \frac{\frac{1}{2}}{1+t} dt + \int \frac{-\frac{1}{2}t + \frac{1}{2}}{1+t^2} dt$$

$$= \frac{1}{2} \ln|1+t| + \frac{1}{4} \int \frac{2t}{t^2+1} dt + \frac{1}{2} \int \frac{dt}{1+t^2}$$

$$= \frac{1}{2} \ln|1+t| - \frac{1}{4} \ln|t^2+1| + \frac{1}{2} \arctan t + C$$

$$= \frac{1}{2} \ln|1+\tan x| - \frac{1}{4} \ln|\tan^2 x + 1| + \frac{1}{2} \arctan(\tan x) + C$$

2. $L(x) = \int \frac{2x}{x^2+x+4} dx$

$\xrightarrow{1/2} \int \frac{2x+1-1}{x^2+x+4} dx$

$\xrightarrow{1/2} \int \frac{2x+1}{x^2+x+4} dx - \int \frac{dx}{x^2+x+4}$

$= \ln|x^2+x+4| - \int \frac{dx}{(x+1/2)^2 + (\sqrt{15}/4)^2}$

$= \ln|x^2+x+4| - \frac{1}{\sqrt{15}/4} \arctan\left(\frac{x+1/2}{\sqrt{15}/4}\right) + C$

$= \ln|x^2+x+4| - \frac{4}{\sqrt{15}} \arctan\left(\frac{2x+1}{\sqrt{15}}\right) + C$

$M(x) = \int \frac{\cos x \sin x}{\cos^2 x + \cos x + 4} dx$

$\left\{ \begin{array}{l} t = \cos x \Rightarrow dt = -\sin x dx \end{array} \right.$

$\xrightarrow{1/2} \int \frac{t}{t^2+t+4} dt = -\frac{1}{2} L(t)$

$\xrightarrow{1/2} -\frac{1}{2} L(\cos x)$

3. $f(x) = \frac{x^2-1}{(x^2-4)^2(x^2+4)^2}$

$\xrightarrow{1/2} \frac{x^2-1}{(x-2)^2(x+2)^2(x^2+4)^2}$

irreducible & $\deg(\text{num}) < \deg(\text{den})$

$= \frac{A}{x-2} + \frac{B}{(x-2)^2} + \frac{C}{x+2} + \frac{D}{(x+2)^2} + \frac{Ex+F}{x^2+4} + \frac{Gx+H}{(x^2+4)^2}$

Exercise 1 (course) [5 points]

Let f and g be two integrable functions on $[a, b]$. If g is of constant sign in $[a, b]$, then there exists a real $K \in [m, M]$, where $m = \inf_{x \in [a, b]} f(x)$ and $M = \sup_{x \in [a, b]} f(x)$, such that

$$\int_a^b f(x)g(x)dx = K \int_a^b g(x)dx.$$

Exercise 2 [10 points]

1. Calculate the integral $\int (1 + \cos x) \tan x dx$.
2. Give the general form of the partial fraction decomposition of $f(x)$ (without calculating the constants) for

$$a) \quad f(x) = \frac{1}{x^3(x^2 + 1)}; \quad b) \quad f(x) = \frac{x + 1}{(x^3 + 1)^2}.$$

3. Calculate $\lim_{n \rightarrow +\infty} u_n$, where

$$u_n = \sum_{k=1}^n \frac{k^2}{n^3 + k^3}.$$

Exercise 3 [6 points]

Let

$$I = \int_0^\pi x (\sin x)^2 dx \quad \text{and} \quad J = \int_0^\pi x (\cos x)^2 dx.$$

1. Calculate $I + J$.
2. Calculate I .
3. Deduce J .

Exercise 4 [9 points]

i) Calculate the following integral :

$$F(x) = \int \frac{x \ln x}{(x^2 + 1)^2} dx.$$

ii) Let $a \in]0, 1[$ and $I(a) = \int_a^1 \frac{x \ln x}{(x^2 + 1)^2} dx$.

- 1) Calculate $I(a)$.
- 2) Calculate the following limits :

$$\lim_{a \rightarrow 1^-} I(a) \quad \text{and} \quad \lim_{a \rightarrow 0^+} I(a).$$



Ex. 1) Suppose $g \geq 0$ (If not take $-g$) and Let

$$J = \int_a^b g(x) dx, \quad I = \int_a^b f(x)g(x) dx. \quad \text{Then, } \forall x \in [a, b],$$

$$m \leq f(x) \leq M \Rightarrow m g(x) \leq f(x)g(x) \leq M g(x)$$

$$\xrightarrow{J \geq 0} m J \leq I \leq M J$$

If $J = 0$, then $I = 0 = k J$ with $k = \frac{m+M}{2} \in [m, M]$.

If $J \neq 0$, then $J > 0$ and $m \leq \frac{I}{J} \leq M$.

Let $k = \frac{I}{J}$, then $k \in [m, M]$ and $I = k J$.

Ex. 2) $w(-u) = w(u)$. put $t = \cos u$. Then $dt = -\sin u du$
and $du = -\frac{dt}{\sin u}$. It follows

$$\begin{aligned} \int (1 + \cos u) \tan u du &= - \int (1 + \cos u) \frac{\sin u}{\cos u} \frac{dt}{\sin u} = - \int \frac{(1+t)}{t} dt \\ &= - \int \frac{dt}{t} - \int dt = -\ln|t| - t + C = -\ln|\cos u| - \cos u + C. \end{aligned}$$

$$2) (a) f(x) = \frac{1}{x^3(x^2+1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x^3} + \frac{Dx+E}{x^2+1}$$

$$(b) f(x) = \frac{x+1}{(x^3+1)^2} = \frac{1}{(x+1)(x^2-x+1)^2} = \frac{A}{x+1} + \frac{B_1x+C_1}{x^2-x+1} + \frac{B_2x+C_2}{(x^2-x+1)^2}$$

$$\begin{aligned} 3) u_n &= \sum_{k=1}^n \frac{k^2}{n^3+k^3} = \sum_{k=1}^n \frac{k^2}{n^3(1+\frac{k^3}{n^3})} = \frac{1}{n} \sum_{k=1}^n \frac{(\frac{k}{n})^2}{1+(\frac{k}{n})^3} \\ &\longrightarrow \int_0^1 \frac{x^2}{1+x^3} dx = \frac{1}{3} \int \frac{d(1+x^3)}{(1+x^3)} = \frac{1}{3} \ln(1+x^3) \Big|_0^1 = \frac{1}{3} \ln 2 \end{aligned}$$

Ex. 3

$$I + J = \int_0^{\pi} x(\cos^2 u + \sin^2 u) du = \int_0^{\pi} x du = \left[\frac{x^2}{2} \right]_0^{\pi} = \frac{\pi^2}{2}$$

$$2) \begin{cases} u = x \\ u' = 1 \end{cases} \quad \begin{cases} v' = (\sin u)^2 = \frac{1 - \cos 2u}{2} \\ v = \frac{x}{2} - \frac{\sin 2u}{4} \end{cases}$$

$$\Rightarrow I = uv \Big|_0^{\pi} - \frac{1}{2} \int_0^{\pi} \left(x - \frac{\sin 2u}{2} \right) du = \frac{\pi^2}{2} - \frac{1}{2} \left[\frac{x^2}{2} + \frac{\cos 2u}{4} \right]_0^{\pi}$$

$$= \frac{\pi^2}{2} - \frac{1}{2} \left[\frac{\pi^2}{2} + \frac{1}{4} - \frac{1}{4} \right] = \frac{\pi^2}{2} - \frac{\pi^2}{4} = \frac{\pi^2}{4}$$

$$3) J = I + J - I = \frac{\pi^2}{2} - \frac{\pi^2}{4} = \frac{\pi^2}{4}$$

Ex. 4

$$\begin{cases} u = \ln x \\ u' = \frac{1}{x} \end{cases} \quad \begin{cases} v' = \frac{x}{(x^2+1)^2} \\ v = \frac{1}{2} \int \frac{d(x^2+1)}{(x^2+1)^2} = -\frac{1}{2(x^2+1)} \end{cases}$$

$$\Rightarrow f(x) = -\frac{\ln x}{2(x^2+1)} + \frac{1}{2} \int \frac{du}{x(x^2+1)} = uv + \frac{1}{2} \int \frac{1+x^2-x^2}{x(x^2+1)} du$$

$$= uv + \frac{1}{2} \int \frac{du}{x} - \frac{1}{2} \int \frac{x}{x^2+1} du = -\frac{1}{2} \frac{\ln x}{(x^2+1)} + \frac{\ln x}{2} - \frac{1}{4} \ln(x^2+1) + C$$

ii)

$$1) I(a) = f(1) - f(a) = -\frac{1}{4} \ln 2 + \frac{1}{2} \frac{\ln a}{a^2+1} - \frac{\ln a}{2} + \frac{1}{4} \ln(a^2+1)$$

$$= -\frac{\ln 2}{4} + \frac{\ln a}{2} \left(\frac{1}{a^2+1} - 1 \right) + \frac{1}{4} \ln(a^2+1)$$

ii)

$$2) \lim_{a \rightarrow 1^-} I(a) = -\frac{\ln 2}{4} + \frac{1}{4} \ln 2 = 0$$

$$\lim_{a \rightarrow 0^+} I(a) = \lim_{a \rightarrow 0^+} \left[-\frac{\ln 2}{4} - \frac{a^2 \ln a}{2(a^2+1)} + \frac{1}{4} \ln(a^2+1) \right]$$

$$= -\frac{\ln 2}{4} - 0 + 0 = -\frac{\ln 2}{4}$$

Exercice 1 (13 points)

1. Soit

$$I = \int_0^{\frac{\pi}{2}} \frac{\sqrt{1 + \cos x}}{\sqrt{1 + \cos x} + \sqrt{1 + \sin x}} dx \text{ et } J = \int_0^{\frac{\pi}{2}} \frac{\sqrt{1 + \sin x}}{\sqrt{1 + \cos x} + \sqrt{1 + \sin x}} dx.$$

Calculer $I + J$ et en déduire les valeurs de I et J .

2 Calculer $\int (\cos x)^3 \sin(\sin x) dx$.

3. Calculer $\int \frac{dt}{(1-t)(1+t)(1+2t)}$, puis $\int \frac{dx}{\sin x + \sin 2x}$.

Exercice 2 (9 points)

Soit $x \in \mathbb{R}$ et f une fonction continue sur $\mathbb{R}$.

a) Montrer que si f est impaire, alors $\int_{-x}^x f(t) dt = 0$.

b) On suppose que f est T -périodique sur $\mathbb{R}$. Soit $F(x) = \int_{x-T}^{x+T} f(t) dt$.

i) Calculer $F'(x)$.

ii) En déduire que $F(x) = \int_{-T}^{+T} f(t) dt$.

iii) Calculer $\int_{3\pi}^{5\pi} \frac{\sin t}{(\sin t)^2 + \cos t + 2} dt$.

Exercice 3 (8 points)

1. Montrer que $\lim_{n \rightarrow +\infty} \int_1^2 \frac{e^{-nt^2}}{1+t} dt = 0$.

2. Soit $h(x) = \int_x^{x^2} \frac{1}{\ln t} dt$.

Calculer les limites suivantes:

$$\lim_{x \rightarrow 0^+} h(x), \quad \lim_{x \rightarrow 1^+} h(x) \text{ et } \lim_{x \rightarrow +\infty} h(x).$$



Bonne chance.

En.1 (1) $I + J = \int_0^{\frac{\pi}{2}} du = \frac{\pi}{2}$

Let $u = \frac{\pi}{2} - n$, then $du = -dn$ and $\begin{cases} \cos u = \sin n \\ \sin u = \cos n \end{cases}$

So $I = - \int_{\frac{\pi}{2}}^0 \frac{\sqrt{1+\sin u}}{\sqrt{1+\sin u} + \sqrt{1+\cos u}} du = \int_0^{\frac{\pi}{2}} \frac{\sqrt{1+\sin u}}{\sqrt{1+\sin u} + \sqrt{1+\cos u}} du$

$= J \Rightarrow 2I = \frac{\pi}{2} \Rightarrow I = \frac{\pi}{4} = J$

(2) $\int (\cos n)^3 \sin(\sin n) dn = \int (\cos u)^2 \sin(\sin u) \cos u du$

$u = \sin n$

$$\int (1-u^2) \sin u du$$

$1-u^2$	}	$\sin u$
$-2u$		$-\cos u$
-2		$-\sin u$
0		$\cos u$

$= -\cos u (1-u^2) - 2u \sin u - 2 \cos u + K$

$= -\cos(\sin u) (\cos u)^2 - 2 \sin u \sin(\sin u)$

$= -2 \cos(\sin u) + K$

(3) $f(t) = \frac{A}{1-t} + \frac{B}{1+t} + \frac{C}{1+2t} = \frac{1}{6(1-t)} - \frac{1}{2(1+t)} + \frac{4}{3(1+2t)}$

$\Rightarrow I(t) = \int f(t) dt = -\frac{1}{6} \ln|1-t| - \frac{1}{2} \ln|1+t| + \frac{2}{3} \ln|1+2t| + K$

$J(u) = \int \frac{du}{\sin u + \sin 2u} = \int \frac{du}{\sin u + 2 \sin u \cos u}$

$= \int \frac{du}{\sin u (1 + 2 \cos u)} = - \int \frac{dt}{(1-t^2)(1+2t)} = -I(\cos u)$

$w(-u) = w(u)$
 Let $t = \cos u$

Ex. 2 (a) $I(n) \stackrel{u=-t}{=} \int_n^{-n} f(-u) (-du) \stackrel{f \text{ odd}}{=} \int_n^{-n} f(u) du$
 $= - \int_n^{-n} f(u) du = -I(n) \Rightarrow 2I(n) = 0 \Rightarrow I(n) = 0$

(b) (i) $f'(n) = f(n+T) - f(n-T) = f(n) - f(n) = 0$

(ii) $f'(n) = 0 \quad \forall n \Rightarrow f(n) = K = f(0) = \int_{-\pi}^{\pi} f(t) dt.$

(iii) $I = \int_{3\pi}^{5\pi} f(t) dt$; f is cont and 2π -periodic over $\mathbb{R}$ $\Rightarrow$

$I = \int_{3\pi-4\pi}^{5\pi-4\pi} f(t) dt = \int_{-\pi}^{\pi} f(t) dt = 0$ since f is odd.

Ex. 3 let $I_n = \int_1^2 \frac{e^{-nt^2}}{1+t} dt = \int_1^2 f_n(t) dt$

$\forall t \in [1, 2] : 0 \leq f_n(t) \leq e^{-nt^2} \leq e^{-n}.$

$\Rightarrow 0 \leq |I_n| = I_n \leq \int_1^2 e^{-n} dt = e^{-n} \xrightarrow{n \rightarrow \infty} 0$

(2.) $h(n) = \int_n^{n^2} \frac{t}{t \ln t} dt = \int_n^{n^2} f(t) g(t) dt$ with $f(t) = t$.

(a) If $n \in]0, \frac{1}{2}[$, $g(n)$ has a const sign < 0 and f is cont.

$h(n) \stackrel{MVT}{=} f(c_n) \int_n^{n^2} \frac{dt}{t \ln t} = c_n \ln |c_n + 1|^{n^2}$ with $c_n \in [n^2, n]$
 $= c_n \ln 2 \xrightarrow[n \rightarrow \infty]{c_n \rightarrow 0} 0$
 (they can write $c_n \in [n, n^2]$)

(b) $h(n) = c_n \ln 2$ with $c_n \in [n, n^2] = [n, n^2]$

$\xrightarrow[n \rightarrow \infty]{c_n \rightarrow 1} \ln 2$

(c) $h(n) = c_n \ln 2$ with $c_n \in [n, n^2]$
 $\Rightarrow n \ln 2 \xrightarrow{n \rightarrow \infty} \infty \Rightarrow \lim_{n \rightarrow \infty} h(n) = \infty$

M. Ziad

Université Libanaise
Faculté des Sciences I
Dépt. Math.

Math 104

Mars, 2014
Examen Partiel
Durée : Une heure

Exercice 1 (13pts) + 1 bonus si + K
Calculer les intégrales suivantes

1. $I(x) = \int e^{2x} \cos(e^x) dx$, $J(x) = \int \frac{x dx}{\sqrt{x^2 + 2x + 2}}$

2. $K(x) = \int \frac{20 dx}{(x-1)(x^2+x-6)}$. En déduire $L(x) = \int \frac{20 \cos x}{\sin^3 x - 7 \sin x + 6} dx$.

3. $M(x) = \int \frac{dx}{(x^2 + 2x + 2)^2}$

Exercice 2 (7pts)

Calculer $\lim_{n \rightarrow \infty} u_n$ dans les cas suivants :

1. $u_n = \prod_{k=0}^{k=n} \left(1 + \frac{k^2}{n^2}\right)^{\frac{1}{n}}$

2. $u_n = \int_2^3 \frac{\sin(nx)}{n \ln x} dx$

Exercice 3 (10pts)

$|u_n| < \frac{1}{n} \int_2^3 \frac{1}{x} dx$

1. Calculer les dérivées de g et h en précisant leurs domaines de définition.

(a) $g(x) = \int_x^{x^2} \frac{\arctan t}{t^2 - 1} dt$

(b) $h(x) = \int_x^{e^x} \frac{\cos t}{t-2} dt$

2. Calculer

$\lim_{x \rightarrow 0^+} \frac{1}{x^2} \int_{\cos x}^{\cos(x^2)} \left(\frac{e^t}{t}\right) dt$

Method 2

$u_n \stackrel{(1pt)}{=} \frac{\sin n C_n}{n}$

$\int_2^3 \frac{dx}{\ln x}$

$\stackrel{(0.5pt)}{=} K$

$\frac{\sin(n C_n)}{n}$

with $C_n \in [2, 3]$

$\stackrel{(0.5)}{\rightarrow} 0$

since $\sin$ is bounded $\stackrel{(0.5pt)}{\rightarrow}$

Exercise I:

Solution Partial Mars 2014

$$I(x) = \int e^{2x} \cos(e^x) dx = \int e^u \cos(e^u) d(e^u)$$

$$\begin{array}{l} u \\ 1 \rightarrow \cos u \\ \quad \rightarrow \sin u \\ 0 \rightarrow -\cos u \end{array}$$

let $u = e^x$ $\Rightarrow \int u \cos u du = u \sin u + \cos u + C$
 $= e^x \sin(e^x) + \cos e^x + C$

For $h(u)$:
 $h(x) = f(u) = \frac{20}{(x-1)(x^2+x-6)} = \frac{20}{(x-1)(x+3)(x-2)} = \frac{A}{x+1} + \frac{B}{x+3} + \frac{C}{x-2}$

$$A = \frac{20}{-4} = -5 ; B = \frac{20}{(-3-1)(-3-2)} = 1 ; C = \frac{20}{5} = 4$$

$$h(x) = -5 \ln|x-1| + \ln|x+3| + 4 \ln|x-2| + C$$

let $u = \sin x$
 $L(x) = \int \frac{20 \cos x}{(\sin^2 x) - 7 \sin x + 6} dx = \int \frac{20 du}{u^2 - 7u + 6} = h(u)$
 $= -5 \ln(1 - \sin x) + \ln(3 + \sin x) + 4 \ln(2 - \sin x)$

$$J(x) = \frac{1}{2} \int \frac{2x dx}{u} ; \text{ let } u = \sqrt{x^2 + 2x + 2}$$

$$= \frac{1}{2} \int \frac{(2x+2-2) dx}{\sqrt{u}} = \frac{1}{2} \int \frac{du}{\sqrt{u}} - \int \frac{du}{\sqrt{u}}$$

$$= \sqrt{u} - \int \frac{du}{\sqrt{(x+1)^2+1}} = \sqrt{x^2+2x+2} - \ln(x+1+\sqrt{(x+1)^2+1}) + C$$

$$M(x) = \int \frac{du}{(x^2+2x+2)^2} = \int \frac{du}{[u^2+1]^2}, \text{ let } u=x+1$$

$$\Rightarrow \int \frac{du}{[u^2+1]^2} = \int \frac{1+u^2-u^2}{(u^2+1)^2} du = \int \frac{du}{1+u^2} - \int u \cdot \frac{u du}{(u^2+1)}$$

$$\text{for } M(x) = \int \frac{du}{(x^2+2x+2)^2} = \int \frac{du}{(u^2+1)^2}, u=x+1$$

$$M_1 = \int \frac{du}{u^2+1} = \arctan u$$

$$M_2 = \int u \cdot \frac{du}{u^2+1} = u \cdot \frac{1}{u^2+1} - \int u \cdot \frac{-2u}{(u^2+1)^2} du = \frac{u}{u^2+1} + 2 \int \frac{u^2}{(u^2+1)^2} du$$

$$= \frac{u}{u^2+1} + 2 \int \frac{u^2+1-1}{(u^2+1)^2} du = \frac{u}{u^2+1} + 2M_1 - 2M_2$$

$$M_2 = \frac{1}{2} \left[M_1 + \frac{u}{u^2+1} \right]$$

$$= \arctan u + \frac{u}{2(u^2+1)} - \frac{1}{2} \int \frac{du}{u^2+1}$$

$$= \frac{1}{2} \arctan u + \frac{u}{2(u^2+1)} + C = \frac{1}{2} \arctan(x+1) + \frac{x+1}{2(x^2+2x+2)} + C$$

+ C

Exercise III :

1) a) $g'(x) = 2n \frac{\arctan n^2}{n^4 - 1} - \frac{\arctan n}{n^2 - 1}$, $D =]-1, +\infty[$

b) $h'(x) = \frac{e^x \cos e^x}{e^x - 2} - \frac{\cos n}{n - 2}$, $D =]-\infty, \ln 2[\cup]2, +\infty[$

2)
$$I_n = \frac{1}{n^2} \int_{\cos n}^{\cos n^2} \frac{e^t}{t} dt \stackrel{MVT}{=} \frac{e^{\xi}}{e^{\eta}} \cdot \frac{1}{n^2} \int_{\cos n}^{\cos n^2} dt$$

$$= \frac{e^{\xi}}{e^{\eta}} \cdot \frac{\cos(n^2) - \cos n}{n^2}$$

$$\stackrel{N(\omega)}{\sim} \frac{e^{\xi}}{e^{\eta}} \frac{\left[1 - \left(1 - \frac{n^2}{2}\right)\right]}{n^2} \xrightarrow[n \rightarrow +\infty]{\xi \rightarrow 1} \frac{e}{2}$$



Exercise II:

$$1) U_n = \prod_{k=0}^n \left(1 + \frac{k^2}{n^2}\right) \Rightarrow \ln U_n = \frac{1}{n} \sum_{k=0}^n \ln\left(1 + \left(\frac{k}{n}\right)^2\right)$$

$$= 0 + \frac{1}{n} \sum_{k=1}^n \ln\left(1 + \left(\frac{k}{n}\right)^2\right) \xrightarrow{n \rightarrow \infty} \int_0^1 \ln(1+x^2) dx = 1$$

$$\begin{array}{l|l} U = \ln(1+x^2) & V' = 1 \\ U' = \frac{2x}{1+x^2} & V = x \end{array} \quad \left| \quad I = x \ln(1+x^2) \right|_0^1 - 2 \int_0^1 \frac{x^2}{1+x^2} dx$$

$$I(x) = \ln 2 - 2 \left[\int_0^1 dx - \arctan x \right]_0^1 = \ln 2 - 2 \left[1 - \frac{\pi}{4} \right]$$

$$\text{Hence } U_n \rightarrow e^{\frac{\pi}{2} + \ln 2 - 2} = \frac{\pi}{2} + \ln 2 - 2$$

$$2) 0 \leq |U_n| \leq \int_2^3 \frac{|\sin(nu)|}{n \ln n} du \leq \frac{1}{n} \int_2^3 \frac{du}{\ln u} \leq \frac{1}{n} \rightarrow 0$$

$$\Rightarrow U_n \xrightarrow{n \rightarrow \infty} 0$$

(73)



Math 104
Partial Exam

Duration : One hour.

Ex.I [15 pts]

Calculate the following integrals :

(i) $\int \frac{dx}{x\sqrt{1+x}}$, (ii) $\int (\sin x) \ln(\cos x) dx$, (iii) $\int \frac{dx}{(\cos x)^5}$.

(iv) $\int \frac{x}{\sqrt{x^2+x+2}} dx$. Deduce $\int \frac{\sin(2x)}{\sqrt{(\cos x)^2 + \cos x + 2}} dx$.

(v) $\int \frac{dx}{2+\cos x}$. Deduce $\int_0^\pi \frac{dx}{2+\cos x}$.

Ex.II [4 pts]

Let $f(x) = \frac{(\cos x)^3 e^{(\sin x)^4}}{(\sin x)^{20} + (\sin x)^2 + 2}$.

(a) Calculate $\int_\pi^{2\pi} f(x) dx$.

(b) Calculate $\int_{-1}^1 x f(x) dx$.

Ex.III [11 pts]

(a) Let $u_n = \sum_{k=1}^n \sqrt{\frac{n^2 - k^2}{n^4}}$ and $v_n = \left(\prod_{k=1}^n \sqrt{\frac{n^2 + k^2}{n^2}} \right)^{\frac{1}{n}}$.

Calculate $\lim_{n \rightarrow \infty} u_n$ and $\lim_{n \rightarrow \infty} v_n$.

(b) Let $g(x) = \int_{\sin x}^{\ln(x+1)} x^3 \sin(e^t) dt$. Calculate $g'(x)$.

(c) Compute the following limits :

$$\lim_{x \rightarrow 1^+} \int_{x^2}^{x^3} \frac{\cos(\frac{\pi}{2}t)}{t \ln t} dt, \quad \lim_{x \rightarrow 0} \frac{1}{x} \int_{\sin x}^{x^3} e^{\sqrt{1-t^2}} dt.$$



Ex. 1:

(i) $\int \frac{dx}{x\sqrt{1+x}}$ $t = \sqrt{1+x}$ $1+x = t^2$ $x = t^2 - 1$
 $dx = 2t dt$ $\int \frac{dx}{x\sqrt{1+x}} = \int \frac{2t dt}{(t^2-1)t} = 2 \int \frac{dt}{t^2-1}$

$= \ln \left| \frac{t-1}{t+1} \right| + C$

(ii) $\int \sin x \ln(\cos x) dx$ $u = \cos x$ $du = -\sin x dx$

$\int \sin x \ln(\cos x) dx = - \int \ln u du = - \int (u)' \ln u du$

$= \int u (\ln u)' du = u \ln u = \int du - u \ln u = u - u \ln u + C$

$= (\cos x - 1) \ln(\cos x) + C$

(iii) $\int \frac{dx}{(\cos x)^5} = \int u v' dx$ $u = \frac{1}{\cos^3 x}$ & $v = \tan x$

$\int \frac{dx}{(\cos x)^5} = \int u v' dx = - \int u' v dx + uv = 3 \int \frac{\sin^2 x}{\cos^5 x} dx + \frac{\tan x}{\cos^3 x}$

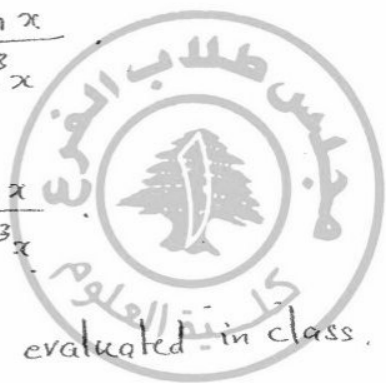
$= 3 \int \frac{1 - \cos^2 x}{\cos^5 x} dx + \frac{\tan x}{\cos^3 x} = + 3 \int \frac{\cos^2 x}{\cos^5 x} dx - 3 \int \frac{dx}{\cos^3 x} + \frac{\tan x}{\cos^3 x}$

Thus,

(1-3) $\int \frac{dx}{(\cos x)^5} = - 3 \int \frac{dx}{\cos^3 x} + \frac{\tan x}{\cos^3 x}$

$\int \frac{dx}{(\cos x)^5} = \frac{3}{2} \int \frac{dx}{\cos^3 x} - \frac{1}{2} \frac{\tan x}{\cos^3 x}$

this one has been evaluated in class.



$$\begin{aligned}
 \int \frac{dx}{\cos^3 x} &= \int \frac{1}{\cos x} \underbrace{\frac{1}{\cos^2 x}}_{= (\tan x)'} dx \\
 &= - \int \left(\frac{1}{\cos x} \right)' \tan x dx + \frac{\tan x}{\cos x} \\
 &= - \int \frac{\sin^2 x}{\cos^3 x} dx + \frac{\tan x}{\cos x} \\
 &= - \int \frac{1 - \cos^2 x}{\cos^3 x} dx + \frac{\tan x}{\cos x} \\
 &= - \int \frac{dx}{\cos^3 x} + \int \frac{dx}{\cos x} + \frac{\tan x}{\cos x}
 \end{aligned}$$

Thus, $\int \frac{dx}{\cos^3 x} = \frac{1}{2} \int \frac{dx}{\cos x} + \frac{\tan x}{2 \cos x}$
 solved in class (use $t = \tan \frac{x}{2}$).

(iv) $\int \frac{x dx}{\sqrt{x^2 + x + 2}}$... $x^2 + x + 2 = \left(x + \frac{1}{2}\right)^2 + \frac{7}{4}$

$x = m \left(x + \frac{1}{2}\right) + n$, $m = 1$ & $n = -\frac{1}{2}$, $u = \left(x + \frac{1}{2}\right)$

Thus,

$$\begin{aligned}
 \int \frac{x dx}{\sqrt{x^2 + x + 2}} &= \int \frac{\left(x + \frac{1}{2}\right)}{\sqrt{\left(x + \frac{1}{2}\right)^2 + \frac{7}{4}}} dx - \frac{1}{2} \int \frac{dx}{\sqrt{\left(x + \frac{1}{2}\right)^2 + \frac{7}{4}}} \\
 &= \int \frac{u du}{\sqrt{u^2 + \frac{7}{4}}} - \frac{1}{2} \int \frac{du}{\sqrt{u^2 + \frac{7}{4}}} \\
 &= \frac{1}{2} \frac{\left(u^2 + \frac{7}{4}\right)^{1-\frac{1}{2}}}{1-\frac{1}{2}} - \frac{1}{2} \ln \left(u + \sqrt{u^2 + \frac{7}{4}}\right) + C
 \end{aligned}$$

(81)

$$\int \frac{\sin(2x)}{\sqrt{\cos^2 x + \cos x + 2}} dx = 2 \int \frac{\cos x \sin x}{\sqrt{\cos^2 x + \cos x + 2}} dx$$

Thus, $t = \cos x, dt = -\sin x dx$

$$\int \frac{\sin(2x) dx}{\sqrt{\cos^2 x + \cos x + 2}} = -2 \int \frac{t dt}{\sqrt{t^2 + t + 2}}$$

$$= -(t^2 + t + 2)^{1/2} - \frac{1}{2} \ln \left(t + \frac{1}{2} + \sqrt{t^2 + t + 2} \right) + C$$

(v) $\int \frac{dx}{2 + \cos x} = F(x)$. Use $t = \tan \frac{x}{2}$. The expression

of $F(x)$ is found in class. $\int_0^{\pi} \frac{dx}{2 + \cos x} = F(\pi) - F(0)$.

Ex. II: $f(x) = \frac{(\cos x)^3 e^{(\sin x)^4}}{(\sin x)^{20} + (\sin x)^2 + 2}$

(a) $u = \sin x, du = \cos x dx$. $\cos^2 x = 1 - \sin^2 x = 1 - u^2$.

$x = \pi \Rightarrow u = 0$ & $x = 2\pi \Rightarrow u = 0$.

$$f(x) dx = \frac{\cos^2 x \cdot e^{(\sin x)^4}}{(\sin x)^{20} + (\sin x)^2 + 2} \cos x dx = \frac{(1 - u^2) e^{u^4}}{u^{20} + u^2 + 2} du$$

$$\int_{\pi}^{2\pi} f(x) dx = \int_0^0 \frac{(1 - u^2) e^{u^4}}{u^{20} + u^2 + 2} du = 0$$

b) $g(x) = x f(x)$. $g(-x) = -x f(-x) = -x f(x) = -g(x)$.

g is odd. $\int_{-1}^1 g(x) dx = 0$.



(82)

Ex. III:

$$(a) \quad u_n = \sum_{k=1}^n \sqrt{\frac{n^2 - k^2}{n^4}} = \sum_{k=1}^n \frac{1}{n^2} \sqrt{n^2 - k^2}$$

$$= \sum_{k=1}^n \frac{1}{n} \sqrt{1 - \left(\frac{k}{n}\right)^2} = \frac{1}{n} \sum_{k=1}^n f\left(\frac{k}{n}\right), \quad f(x) = \sqrt{1 - x^2}$$

$$\lim_{n \rightarrow \infty} u_n = \int_0^1 \sqrt{1 - x^2} dx = \dots$$

(b)

$$\ln v_n = \frac{1}{n} \sum_{k=1}^n \ln \sqrt{\frac{n^2 + k^2}{n^2}}$$

$$= \frac{1}{n} \sum_{k=1}^n \frac{1}{2} \ln \left(\frac{n^2 + k^2}{n^2} \right)$$

$$= \frac{1}{2} \frac{1}{n} \sum_{k=1}^n \ln \left(1 + \left(\frac{k}{n}\right)^2 \right) = \frac{1}{2} \frac{1}{n} \sum_{k=1}^n f\left(\frac{k}{n}\right),$$

$$f(x) = \ln(1 + x^2)$$

$$\lim_{n \rightarrow \infty} \ln v_n = \frac{1}{2} \int_0^1 \ln(1 + x^2) dx = \dots$$

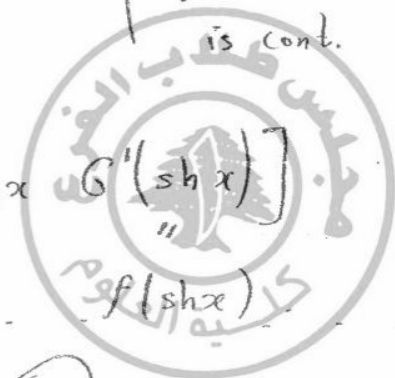
$$\lim_{n \rightarrow \infty} v_n = \exp \left(\frac{1}{2} \int_0^1 \ln(1 + x^2) dx \right)$$

$$(b) \quad g(x) = x^3 \left[G(\ln(x+1)) - G(\operatorname{sh} x) \right] \quad \left| \quad \begin{array}{l} G(u) = \int_0^u f(t) dt \\ f(t) = \sin(e^t) \end{array} \right.$$

$$g'(x) = 3x^2 \left[G(\ln(x+1)) - G(\operatorname{sh} x) \right]$$

$$+ x^3 \left[\frac{1}{x+1} G'(\ln(x+1)) - \operatorname{ch} x G'(\operatorname{sh} x) \right]$$

$$f(\ln(x+1)) \quad \&$$



(83)

$$(c) \int_{x^2}^{\frac{\cos(\frac{\pi}{2}c)}{t \ln t}} dz = \int_{x^2}^{\frac{\cos(\frac{\pi}{2}c)}{t \ln t}} \frac{dt}{t \ln t}$$

$$= \frac{\cos(\frac{\pi}{2}c)}{\ln c} \left[\frac{-1}{(\ln t)^2} \right]_{t=x^2}^{t=x^3}$$

$$= -\frac{5}{36} \cos\left(\frac{\pi}{2}c\right) \frac{1}{\ln x}, \quad \text{where } x^2 \leq c \leq x^3.$$

As $x \rightarrow 1_+$, $c \rightarrow 1$, and

$$\begin{aligned} \cos\left(\frac{\pi}{2}c\right) &= \cos\left(\frac{\pi}{2}(c-1) + \frac{\pi}{2}\right) \\ &= -\sin\left(\frac{\pi}{2}(c-1)\right) \sim \frac{\pi}{2}(c-1) \quad (c \rightarrow 1_+) \end{aligned}$$

and

$$\frac{\cos(\frac{\pi}{2}c)}{\ln x} \sim \frac{\pi}{2} \cdot \frac{c-1}{\ln x} \in \left[\frac{\pi}{2} \cdot \frac{x^2-1}{\ln x}, \frac{\pi}{2} \cdot \frac{x^3-1}{\ln x} \right]$$

Since $\lim_{x \rightarrow 1_+} \frac{x^2-1}{\ln x} = \lim_{x \rightarrow 1_+} \frac{x^3-1}{\ln x} = 0$, we

deduce that

$$\frac{c-1}{\ln x} \xrightarrow{x \rightarrow 1_+} 0 \quad \& \quad \frac{\cos(\frac{\pi}{2}c)}{\ln x} \xrightarrow{x \rightarrow 1_+} 0.$$

$$\frac{1}{x} \int_{\sin x}^{x^3} e^{\sqrt{1+t}} dt = \frac{F(x^3) - F(0)}{x - 0} = \frac{F(\sin x) - F(0)}{x - 0}$$

$$\begin{aligned} \frac{F(x)}{x} &= \int_0^x e^{\sqrt{1+t}} dt \\ &\xrightarrow{x \rightarrow 0} \frac{d}{dx} F(x^3) \Big|_{x=0} = \frac{d}{dx} F(\sin x) \Big|_{x=0} \\ &= 3x^2 F'(x^3) \Big|_{x=0} = \cos x F'(\sin x) \Big|_{x=0} = \sqrt{1+0} \end{aligned}$$

Math 104
Partial Exam

Duration : One hour.

EX.I [17 pts]

1. Calculate the following integrals :

$$A = \int \frac{dt}{t(t^2 + 2t + 2)}; \quad B = \int \frac{1-x}{\sqrt{2x^2 + 6x - 3}} dx; \quad C = \int_0^{\sqrt{3}} \sqrt{3-x^2} dx.$$

2. (a) Compute $\int \frac{dt}{t^2(1+t)}$.

(b) Deduce $\int \frac{\sin x}{\cos^2 x(1+\cos x)} dx$

(c) Calculate $\int \frac{\sin x}{\cos^3 x} \ln(1+\cos x) dx$.

EX.II [08 pts]

Let $F(x) = \int_{v(x)}^{u(x)} f(t) dt$ and $G(t) = \int f(t) dt$, where f is a continuous function over $[a, b]$, u and v are differentiable with values in $[a, b]$.

1. Find $F'(x)$ in terms of $G(x)$.

2. Compute $F''(x)$.

3. Let $g(x) = \ln(x) \int_{\sin x}^{\tan x} \frac{dt}{1+(\ln t)^2}$. Compute $g'(x)$.

EX.III [05 pts]

We admit that if f is a continuous function and has a constant sign over $[a, b]$ and if $\int_a^b f(x) dx = 0$ then $f = 0$ on $[a, b]$.

Let $f : [0, 1] \rightarrow [0, 1]$ be a continuous function.

1. Show that $f(x)(1-f(x)) \leq 0$ for every $x \in [0, 1]$.

2. Suppose that $f \neq 0$ and

$$\int_0^1 f(x) dx = \int_0^1 f^2(x) dx.$$

Show that $f(x) = 1$ for every $x \in [0, 1]$.



Math 09 - partial 2010

$$Ex 1) A = \int \frac{dt}{t(t^2+2t+2)}$$

Since $t^2+2t+2 = (t+1)^2 + 1$ has no roots, we decompose the integral:

$$\frac{a}{t} + \frac{bt+c}{t^2+2t+2} = \frac{a(t^2+2t+2) + (bt+c)t}{t(t^2+2t+2)}$$

$$= \frac{(a+b)t^2 + (2a+c)t + 2a}{t(t^2+2t+2)}$$

By comparison: $a+b=0$ $2a+c=0$ $2a=1$

$b=-\frac{1}{2}$ $c=-1$ $a=\frac{1}{2}$

$$A = \frac{1}{2} \int \frac{dt}{t} - \int \frac{\frac{1}{2}t-1}{t^2+2t+2} dt$$

$$= \frac{1}{2} \ln|t| - \frac{1}{2} \int \frac{t}{t^2+2t+2} dt - \int \frac{1}{t^2+2t+2} dt$$

$$= \frac{1}{2} \ln|t| - \frac{1}{4} \int \frac{2t+2-2}{t^2+2t+2} dt - \frac{1}{2} \int \frac{dt}{t^2+2t+2}$$

$$= \frac{1}{2} \ln|t| - \frac{1}{4} \int \frac{2t+2}{t^2+2t+2} dt - \frac{1}{4} \ln|1^2+2t+2| + C$$

(86)



$$B = \int \frac{1-x}{\sqrt{2x^2+6x-3}} dx$$

$$= \frac{1}{\sqrt{2}} \int \frac{1-x}{\sqrt{x^2+3x-3/2}} dx = \frac{1}{\sqrt{2}} \int \frac{1-x}{\sqrt{\left(x+\frac{3}{2}\right)^2 - \frac{9}{4} - \frac{3}{2}}} dx$$

$$= \frac{1}{\sqrt{2}} \int \frac{1-x}{\sqrt{\left(x+\frac{3}{2}\right)^2 - \frac{15}{4}}} dx = \frac{2}{\sqrt{2}\sqrt{15}} \int \frac{1-x}{\sqrt{\left[\frac{2}{\sqrt{15}}\left(x+\frac{3}{2}\right)\right]^2 - 1}} dx$$

$$= \frac{2}{\sqrt{2}\sqrt{15}} \int \frac{1 - \left(x + \frac{3}{2}\right) + \frac{3}{2}}{\sqrt{\left[\frac{2}{\sqrt{15}}\left(x + \frac{3}{2}\right)\right]^2 - 1}} dx$$

$$= \frac{-2}{\sqrt{2}\sqrt{15}} \int \frac{x + \frac{3}{2}}{\sqrt{\left[\frac{2}{\sqrt{15}}\left(x + \frac{3}{2}\right)\right]^2 - 1}} dx + \frac{2}{2\sqrt{2}\sqrt{15}} \int \frac{dx}{\sqrt{\quad}}$$

the integral is

$u = \frac{2}{\sqrt{15}}\left(x + \frac{3}{2}\right)$
& the integral is

$$+ \frac{1}{2\sqrt{2}} \int \frac{du}{\sqrt{u^2 - 1}}$$

$$= -\frac{1}{\sqrt{2}} \sqrt{\left[\frac{2}{\sqrt{15}}\left(x + \frac{3}{2}\right)\right]^2 - 1}$$

$$+ \frac{1}{2\sqrt{2}} \cosh^{-1} u + C$$

$$+ \frac{1}{2\sqrt{2}} \cosh^{-1} \left(\frac{2}{\sqrt{15}} \left(x + \frac{3}{2} \right) \right) + C$$

$$= -\frac{1}{\sqrt{2}} \sqrt{\quad}$$

(87)

77

$$C = \int_0^{\sqrt{3}} \sqrt{3-x^2} dx$$

$$\theta = \arcsin \frac{x}{\sqrt{3}}$$

Let $x = \sqrt{3} \sin \theta$

$$\sqrt{3-x^2} = \sqrt{3} \cos \theta$$

$$dx = \sqrt{3} \cos \theta$$

$$\pi/2$$

$$\int_0^{\sqrt{3}} \sqrt{3-x^2} dx = \int_0^{\pi/2} 3 \cos^2 \theta d\theta$$

$$= 3 \int_0^{\pi/2} \frac{1+\cos 2\theta}{2} d\theta$$

$$= \frac{3\theta}{2} + \frac{3\sin 2\theta}{4} \Big|_{\theta=0}^{\theta=\pi/2}$$

$$= \frac{3\pi}{4}$$

$$2. (a) \int \frac{dt}{t^2(1+t)} = + \int \frac{1}{t} dt + \int \frac{1}{t^2} dt + \int \frac{dt}{1+t} = \ln|t| - \frac{1}{t} + \ln|1+t| + C$$

$$\frac{1}{t^2(t+1)} = \frac{a}{t} + \frac{b}{t^2} + \frac{c}{t+1}$$

$$= \frac{a t(t+1) + c t^2 + b(t+1)}{t^2(t+1)}$$

$$= \frac{(a+c)t^2 + (a+b)t + b}{t^2(t+1)}$$

$$\begin{cases} a+c=0 \\ a+b=0 \\ b=1 \end{cases} \quad \begin{cases} c=-1 \\ a=-1 \\ b=1 \end{cases}$$



(b) Let $t = \cos x$. $dt = -\sin x dx$

$$\int \frac{\sin x}{\cos^2 x (1 + \cos x)} dx = - \int \frac{dt}{t^2 (1+t)}$$

$$= - \ln|t| + \frac{1}{t} - \ln|1+t| + C$$

$$= - \ln|\cos x| + \frac{1}{\cos x} - \ln|1 + \cos x| + C$$

(c) Let $t = \cos x$. $dt = -\sin x dx$.

$$\int \frac{\sin x}{\cos^3 x} \ln(1 + \cos x) dx = - \int \frac{1}{t^3} \ln(1+t) dt$$

$$= \frac{3}{2} \int \left(\frac{1}{t^2}\right)' \ln(1+t) dt = -3 \int \frac{1}{t^2} \frac{1}{1+t} dt$$

integration by parts

$$+ \frac{3}{t^2} \ln(1+t) + C$$

$$= -3 \int \frac{dt}{t^2(1+t)} + \frac{3}{t^2} \ln(1+t) + C$$

$$= -3 \ln|t| + \frac{3}{t} - 3 \ln|1+t| + \frac{3}{t^2} \ln(1+t) + C$$



Ex. II :

$$1. F(x) = G(u(x)) - G(v(x)) + C.$$

$$2. F'(x) = u'(x) G'(u(x)) - v'(x) G'(v(x)) \\ = u'(x) f(u(x)) - v'(x) f(v(x)).$$

$$3. g(x) = \ln x \int_{\sin x}^{\tan x} \frac{dt}{1 + (\ln t)^2}.$$

$$= \ln x \cdot F(x), \quad \begin{aligned} v(x) &= \sin x \\ u(x) &= \tan x. \end{aligned}$$

$$g'(x) = \frac{1}{x} F(x) + \ln x \cdot F'(x) \\ = \frac{1}{x} \int_{\sin x}^{\tan x} \frac{dt}{1 + (\ln t)^2} + \frac{(\ln x)}{\cos^2 x} \cdot \frac{1}{1 + (\ln(\tan x))^2}$$

$$- \frac{(\ln x) \cos x}{1 + (\ln(\sin x))^2}.$$

Ex. III :

$$1. f(x) (1 - f(x)) \geq 0 \\ 0 \leq 1 - f(x) \leq 1$$

$$0 \leq f(x) \leq 1$$

$$0 \leq f(x) (1 - f(x)) \leq 1$$

$$2. f \neq 0: \quad \int_0^1 f(x) dx = \int_0^1 [f(x)]^2 dx.$$

$$\int_0^1 [f(x) - (f(x))^2] dx = 0 \Rightarrow \int_0^1 f(x) [1 - f(x)] dx = 0 \Rightarrow f(x) = f(x) [1 - f(x)]$$

$$\text{and hence } \int_0^1 h(x) dx = 0 \Rightarrow \text{hence } h(x) = 0 \quad \forall x \in [0, 1]$$

Math 105
Examen partiel

Durée de l'épreuve : 1 heure.

I. [5 pts] (Cours)

Prouver que si f , u et u' sont continues sur un intervalle ouvert I , de $\mathbb{R}$, contenant $[a, b]$, alors

$$\int_a^b f(u(t))u'(t)dt = \int_{u(a)}^{u(b)} f(u)du.$$

II. [11 pts]

a) Soit f une fonction continue sur $\mathbb{R}$.

i) Déterminer $\lim_{x \rightarrow 0} \frac{1}{x} \int_{-x}^x f(t)dt$.

ii) Soit $g(x) = \int_0^{x \ln x} f(t) \cos(x)dt$. Calculer $g'(x)$.

b) Déterminer les limites suivantes :

i) $\lim_{x \rightarrow \infty} \int_x^{2x} \frac{e^t}{t} dt$ ii) $\lim_{x \rightarrow \infty} \int_x^{2x} \frac{\cos t}{t} dt$.

III. [14 pts]

Calculer les intégrales suivantes :

a)

$$\int_0^{\frac{\pi}{2}} \frac{dx}{4 + \sin x}, \quad \int_{-1}^1 (\cot x + \sqrt{1-x^2})dx.$$

b)

$$\int \frac{x^7}{1+x^{16}} dx, \quad \int \frac{dx}{\sqrt{3x^2+6x+9}}, \quad \int \cos(2x) \ln(\tan x) dx.$$

c)

$$\int \frac{dx}{(\tan x)^6}, \quad \int \frac{dx}{(\operatorname{ch} x)^3}.$$



Ex 1: Question de cours.

Ex 2: a) f est continue sur $\mathbb{R}$.

$$i) \lim_{x \rightarrow 0} \frac{1}{x} \int_{-x}^x f(t) dt ?$$

On a f est continue sur $\mathbb{R}$, en particulier elle est continue sur $[-x, x]$, donc d'après le théorème de la moyenne on obtient que:

$$\exists c \in [-x, x] / \int_{-x}^x f(t) dt = (x - (-x)) f(c)$$

$$\Rightarrow \int_{-x}^x f(t) dt = 2x f(c) \Rightarrow \frac{1}{x} \int_{-x}^x f(t) dt = 2 f(c)$$

or $c \in [-x, x]$, donc si $x \rightarrow 0$ alors $c \rightarrow 0$, d'où:

$$\lim_{x \rightarrow 0} \frac{1}{x} \int_{-x}^x f(t) dt = 2 \lim_{c \rightarrow 0} f(c) = 2 f(0) \text{ car } f \text{ est continue sur } \mathbb{R}.$$

$$ii) g(x) = \int_0^{\sin x} f(t) \cos t dt, \quad g'(x) = ?$$

$$\text{Posons } F(x) = \int_0^x f(t) \cos t dt. \text{ On a } g(x) = F(\sin x).$$

$$\text{On sait que } F'(x) = f(x) \cos x \text{ car } t \rightarrow f(t) \cos t \text{ est continue sur } \mathbb{R}$$

or $x \rightarrow \sin x$ est dérivable sur $\mathbb{R}$, donc:

$$g'(x) = (F(\sin x))' = (\sin x)' F'(\sin x) = \cos x f(\sin x) \cos(\sin x)$$

$$b) i) \lim_{x \rightarrow +\infty} \int_x^{2x} \frac{e^{\frac{1}{t}}}{t} dt.$$

$x \rightarrow +\infty \Rightarrow$ on peut supposer que $x > 0$. On sait que $t \rightarrow \frac{1}{t}$ garde une signe constante sur $]0, +\infty[$ ($\frac{1}{t} > 0$ sur $]0, +\infty[$).

De plus, $t \rightarrow e^{\frac{1}{t}}$ et $t \rightarrow \frac{1}{t}$ sont continue sur $[x, 2x]$, $x > 0$.
D'après le th de la moyenne, on a donc:

$$\int_x^{2x} \frac{e^{\frac{1}{t}}}{t} dt = e^{\frac{1}{c}} \int_x^{2x} \frac{dt}{t} \text{ avec } c \in [x, 2x].$$

$$= e^{\frac{1}{c}} [\ln t]_x^{2x} = e^{\frac{1}{c}} [\ln 2x - \ln x] = e^{\frac{1}{c}} \ln \frac{2x}{x} = e^{\frac{1}{c}} \ln 2.$$

Or $c \in [x, 2x]$ et $x \rightarrow +\infty \Rightarrow c \rightarrow +\infty$, on obtient donc:

$$\lim_{x \rightarrow +\infty} \int_x^{2x} \frac{e^{\frac{1}{t}}}{t} dt = \lim_{c \rightarrow +\infty} \ln e^{\frac{1}{c}} = \ln 2.$$

Intégrons par partie en posant

$$u' = \cos t \quad \text{et} \quad v = \frac{1}{t}.$$

$$\Rightarrow u = \sin t \quad \text{et} \quad v' = -\frac{1}{t^2}.$$

$$\Rightarrow I = \int_n^{2n} \frac{\cos t}{t} dt = \left[\frac{\sin t}{t} \right]_n^{2n} + \int_n^{2n} \frac{\sin t}{t^2} dt$$

$$|I| = \frac{\sin 2n}{2n} - \frac{\sin n}{n} + \int_n^{2n} \frac{\sin t}{t^2} dt$$

$$\lim_{n \rightarrow +\infty} |I(n)| \leq \frac{1}{2|n|} - \frac{1}{|n|} + \frac{1}{2n} - \frac{1}{n}.$$

$$\lim_{n \rightarrow +\infty} |I(n)| \leq \lim_{n \rightarrow +\infty} \left(\frac{1}{2|n|} - \frac{1}{|n|} + \frac{1}{2n} - \frac{1}{n} \right).$$

$$\lim_{n \rightarrow +\infty} |I(n)| \leq 0.$$

Ex III Calculer les intégrales suivantes

$$a) I = \int_0^{\frac{\pi}{2}} \frac{dx}{4 + \sin x}$$

Posons $w(x) = \frac{dx}{4 + \sin x}$, Les trois règles de Bioches ne s'appliquent

pas donc on pose $t = \tan \frac{x}{2}$ (Rq: On a $0 < x < \frac{\pi}{2} \Rightarrow x = 2 \arctan t$ est bien définie)

$$\Rightarrow dx = \frac{2}{1+t^2} dt$$

$$\text{or } \frac{1}{4 + \sin x} = \frac{1}{4 + \frac{2t}{1+t^2}} = \frac{1+t^2}{4t^2 + 2t + 4}, \text{ donc:}$$

$$I = 2 \int_{t(0)}^{t(\frac{\pi}{2})} \frac{dt}{4t^2 + 2t + 4} = 2 \int_0^1 \frac{dt}{4t^2 + 2t + 4}$$

$\frac{1}{4t^2 + 2t + 4}$ est un élément simple du 2^{ème} espèce car le polynôme $4t^2 + 2t + 4$ n'admet pas des racines dans $\mathbb{R}$. 49

Donc, on écrit:

$$4t^2 + 2t + 4 = \left[(2t)^2 + 2t + \left(\frac{1}{4}\right)^2 \right] - \left(\frac{1}{4}\right)^2 + 4$$

$$= \left[\left(2t + \frac{1}{4}\right)^2 + \frac{63}{16} \right] = \frac{63}{16} \left[1 + \left(\frac{8t+1}{\sqrt{63}}\right)^2 \right]$$

$$\text{d'où } I = \frac{16}{63} \times 2 \int_0^1 \frac{dt}{1 + \left(\frac{8t+1}{\sqrt{63}}\right)^2} = \frac{4\sqrt{63}}{63} \int_0^1 \frac{\left(\frac{8t+1}{\sqrt{63}}\right)' dt}{1 + \left(\frac{8t+1}{\sqrt{63}}\right)^2}$$

$$= \frac{4\sqrt{63}}{63} \operatorname{arctg} \left(\frac{8t+1}{\sqrt{63}} \right) \Big|_0^1 = \frac{4\sqrt{63}}{63} \left[\operatorname{arctg} \frac{9}{\sqrt{63}} - \operatorname{arctg} \frac{1}{\sqrt{63}} \right]$$

$$* J = \int_{-1}^1 (\operatorname{catg} x + \sqrt{1-x^2}) dx = \underbrace{\int_{-1}^1 \operatorname{catg} x dx}_{J_1} + \underbrace{\int_{-1}^1 \sqrt{1-x^2} dx}_{J_2}$$

(NB: c'est hors du programme du partiel 2010)

$J_1 = \int_{-1}^1 \operatorname{catg} x dx$. On sait que $t \rightarrow \operatorname{catg} t$ n'est pas définie au point 0, pour cela on coupe J_1 en 0:

$$J_1 = \int_{-1}^0 \operatorname{catg} x dx + \int_0^1 \operatorname{catg} x dx$$

$$= \lim_{a \rightarrow 0} \int_{-1}^a \operatorname{catg} x dx + \lim_{b \rightarrow 0} \int_b^1 \operatorname{catg} x dx$$

$$\text{car: } \int_{-1}^a \operatorname{catg} x dx = \int_{-1}^a \frac{\cos x}{\sin x} dx = \int_{-1}^a \frac{d(\sin x)}{\sin x} = \ln|\sin x| \Big|_{-1}^a$$

$$= \ln|\sin a| - \ln|\sin 1|$$

$$\text{De même } \int_b^1 \operatorname{catg} x dx = \ln|\sin 1| - \ln|\sin b|$$

$$\text{d'où } J_1 = \lim_{a \rightarrow 0} \ln|\sin a| - \lim_{b \rightarrow 0} \ln|\sin b|$$

$$= \lim_{a \rightarrow 0^+} \ln a - \lim_{b \rightarrow 0^+} \ln b = \lim_{(a,b) \rightarrow (0,0)} \ln \frac{a}{b}$$

Si on prend le chemin $a = \alpha b$, $\alpha \in \mathbb{R}^+$:

$$J_1 = \lim_{\substack{(a,b) \rightarrow (0,0) \\ a = \alpha b}} \ln \frac{\alpha b}{b} = \ln \alpha, \quad \alpha \in \mathbb{R}^+ \text{ arbitraire}$$

donc J_1 n'existe pas et par suite J n'existe pas.

$$b) K = \int \frac{x^7}{1+x^8} dx = \int \frac{x^7}{1+(x^8)^2} dx$$

On pose $u = x^8 \Rightarrow du = 8x^7 dx \Rightarrow \frac{du}{8} = x^7 dx$

d'où $K = \frac{1}{8} \int \frac{du}{1+u^2} = \frac{1}{8} \operatorname{arctg} u + C = \frac{1}{8} \operatorname{arctg} x^8 + C$

$$* L = \int \frac{dx}{\sqrt{3x^2+6x+9}} = \frac{1}{\sqrt{3}} \int \frac{dx}{\sqrt{x^2+2x+3}}$$

or $x^2+2x+3 = (x^2+2x+1) + 2 = (x+1)^2 + (\sqrt{2})^2$, de la forme

K^2+u^2 avec $K=\sqrt{2}$ et $u=x+1$, donc:

on pose $u = K \operatorname{sh} t$ c'd d $x+1 = \sqrt{2} \operatorname{sh} t \Rightarrow \begin{cases} dx = \sqrt{2} \operatorname{sh} t \\ \text{et } t = \operatorname{argsh} \left(\frac{x+1}{\sqrt{2}} \right) \end{cases}$

et on a donc:

$x^2+2x+3 = 2 [(\operatorname{sh}^2 t) + 1]$, il résulte que:

$$L = \frac{1}{\sqrt{3}} \int \frac{\operatorname{sh} t}{\sqrt{4 + \operatorname{sh}^2 t}} dt = \frac{1}{\sqrt{3}} \int \frac{\operatorname{sh} t}{\operatorname{ch} t} dt = \frac{1}{\sqrt{3}} \int \frac{d(\operatorname{ch} t)}{\operatorname{ch} t}$$

$$= \frac{1}{\sqrt{3}} \operatorname{Ln} \operatorname{ch} t + C = \frac{1}{\sqrt{3}} \operatorname{Ln} \left[\operatorname{ch} \left(\operatorname{argsh} \left(\frac{x+1}{\sqrt{2}} \right) \right) \right] + C$$

$$* M = \int \cos 2x \operatorname{Ln}(\operatorname{tg} x) dx$$

Intégrons par partie: en posant: $\begin{cases} u = \operatorname{Ln}(\operatorname{tg} x) \Rightarrow u' = \frac{1}{\operatorname{tg} x} = \frac{1}{\cos x \sin x} \\ v' = \cos 2x \Rightarrow v = \frac{1}{2} \sin 2x = \cos x \sin x \end{cases}$

$$M = \int u v' dx = uv - \int u' v dx$$

$$= \frac{1}{2} \sin 2x \operatorname{Ln}(\operatorname{tg} x)$$

$$- \int dx = \frac{1}{2} \sin 2x \operatorname{Ln}(\operatorname{tg} x) - x + C$$

c) $N = \int \frac{dx}{(\operatorname{tg} x)^6}$, Posons $w(x) = \frac{dx}{\operatorname{tg}^6 x}$, on a $w(\pi+x) = w(x)$

$\Rightarrow$ on pose $t = \operatorname{tg} x \Rightarrow x = \operatorname{arctg} t$ avec $t \in \mathbb{R}$ et $-\frac{\pi}{2} < x < \frac{\pi}{2}$

Bioche

$$\Rightarrow dx = \frac{dt}{1+t^2}, \text{ d'où } N = \int \frac{dt}{t^6(1+t^2)}$$

on décompose en élément simple la fraction $\frac{1}{t^6(1+t^2)}$, E utilisant l'une des 3 méthodes suivantes:

1^{ère} méthode:

$$F(t) = \frac{1}{t^6(1+t^2)} = \frac{A}{t} + \frac{B}{t^2} + \frac{C}{t^3} + \frac{D}{t^4} + \frac{E}{t^5} + \frac{F}{t^6} + \frac{Gt+H}{1+t^2}$$

Remarquons que F est une fonction paire, donc le fait que

$F(-t) = F(t)$ nous donne par identification que:

$$A = C = E = G = 0$$

$$\Rightarrow F(t) = \frac{1}{t^6(1+t^2)} = \frac{B}{t^2} + \frac{D}{t^4} + \frac{F}{t^6} + \frac{H}{1+t^2} (*)$$

Multiplications (*) par t^6 et faisons $t=0 \Rightarrow F=1$
 " " " t^2 " " $t \rightarrow \infty \Rightarrow B+H=0 \Rightarrow \boxed{B=-H} \text{ (1)}$

prenons 2 valeurs particulières:

$$t=1 \Rightarrow \boxed{B+2D=-1} \text{ (2)} ; t=2 \Rightarrow \boxed{16B+20D=-4} \text{ (3)}$$

(1), (2) et (3) nous donnent $B=1$, $D=-1$ et $H=-1$, d'où:

$$F(t) = \frac{1}{t^2} + \frac{1}{t^4} - \frac{1}{1+t^2} \text{ et on obtient que:}$$

$$W = -\frac{1}{t} + \frac{1}{3t^3} - \frac{1}{5t^5} - \arctg t + C$$

$$= -\frac{1}{t \operatorname{tg} x} + \frac{1}{3 t^3 \operatorname{tg} x} - \frac{1}{5 t^5 \operatorname{tg} x} - \arctg(\operatorname{tg} x) + C$$

$$= -\frac{1}{t \operatorname{tg} x} + \frac{1}{3 t^3 \operatorname{tg} x} - \frac{1}{5 t^5 \operatorname{tg} x} - x + C$$

2^{ème} méthode: $\frac{1}{t^6(1+t^2)}$ est de type $\frac{1}{t^n(K+t^m)}$, $n \in \mathbb{N}^*$, $m \in \mathbb{N}^*$ d'où $\alpha \in \mathbb{R}$

Avec ce type des fractions, il est beaucoup plus simple, plus rapide
 en éléments simples, d'utiliser ce principe:
 rajouter et retrancher $\frac{t^m}{\alpha}$.

En Effet,

$$\frac{1}{t^6(1+t^2)} = \frac{(1+t^2)-t^2}{t^6(1+t^2)} = \frac{1}{t^6} - \frac{1}{t^4(1+t^2)} = \frac{1}{t^6} - \frac{(1+t^2)-t^2}{(1+t^2)t^4}$$

$$= \frac{1}{t^6} - \frac{1}{t^4} + \frac{1}{t^2(1+t^2)} = \frac{1}{t^6} - \frac{1}{t^4} + \frac{(1+t^2)-t^2}{t^2(1+t^2)}$$

$$= \frac{1}{t^6} - \frac{1}{t^4} + \frac{1}{t^2} - \frac{1}{1+t^2}$$

3^{ème} méthode: On peut utiliser aussi la division successive des puissances
 croissantes de 1 par $1+t^2$, ~~puis~~ ~~de~~ ~~1~~ ~~par~~ ~~1+t^2~~ ~~et~~
 dès qu'on obtient une reste de degré ≤ 6 .

$$\begin{array}{r|l}
 1 & 1+t^2 \\
 -1+t^2 & 1-t^2+t^4 \\
 \hline
 -t^2 & \\
 -t^2-t^4 & \\
 \hline
 t^4 & \\
 t^4+t^6 & \\
 \hline
 -t^6 &
 \end{array}$$

$$\begin{aligned}
 &\Rightarrow 1 = (1+t^2)(1-t^2+t^4) - t^6 \\
 &\Rightarrow \frac{1}{(1+t^2)t^6} = \frac{1-t^2+t^4}{t^6} - \frac{t^6}{t^6(1+t^2)} \\
 &= \frac{1}{t^6} - \frac{1}{t^4} + \frac{1}{t^2} - \frac{1}{1+t^2}
 \end{aligned}$$

$$* K = \int \frac{dx}{(\cosh x)^3}$$

$$\text{on a } K = \int \frac{dx}{\left(\frac{e^x + e^{-x}}{2}\right)^2} = 8 \int \frac{e^{3x}}{(e^{2x} + 1)^3} dx$$

$$\text{on pose } t = e^x \Rightarrow dt = e^x dx \Rightarrow dx = \frac{dt}{t}, \text{ donc:}$$

$$K = 8 \int \frac{t^3}{t(1+t^2)^3} dt = 8 \int \frac{t^2}{(1+t^2)^3} dt$$

On décompose $\frac{t^2}{(1+t^2)^3}$ en éléments simples:

$$F(t) = \frac{t^2}{(1+t^2)^3} = \frac{At+B}{1+t^2} + \frac{Ct+D}{(1+t^2)^2} + \frac{Et+F}{(1+t^2)^3}$$

Remarquons que, $F(-t) = F(t)$ (Fonction paire) donc:

$$F(t) = -\frac{At+B}{1+t^2} + \frac{-Ct+D}{(1+t^2)^2} + \frac{-Et+F}{(1+t^2)^3}, \text{ Par identification on obtient:}$$

$$-A = A, -C = C, -E = E \Rightarrow A = C = E = 0$$

$$\text{d'où } F(t) = \frac{B}{1+t^2} + \frac{D}{(1+t^2)^2} + \frac{F}{(1+t^2)^3} (*)$$

* multiplions (*) par t^2 et faisons $t \rightarrow \infty \Rightarrow 0 = B$

* prenons $t=0$ comme v.p. $\Rightarrow 0 = D + F \Rightarrow D = -F$

$$\text{prenons } t=1 \Rightarrow \frac{1}{8} = \frac{1}{4}D + \frac{1}{8}F \Rightarrow F + 2D = 1$$

d'où $D=1$ et $F=-1$ et on a donc:

$$F(t) = \frac{1}{(1+t^2)^2} + \frac{1}{(1+t^2)^3} \text{ et } F(t) = 8 \int \frac{dt}{(1+t^2)^2} - 8 \int \frac{dt}{(1+t^2)^3}$$

Pour le reste, voir la feuille 1 d'exercice.

Autre méthode: Il est plus efficace d'utiliser la méthode suivante
avec ce type d'intégrale: